Case 5 — Mr. Leonard Vance

66 years old · Male Parkinson's Disease · Diagnosed 12 months ago Hoehn & Yahr Stage I–II (Early) Resting tremor R hand · Micrographia · Hypokinetic gait

Neurological Physiotherapy

■ SUBJECTIVE PROBLEM LIST

- Physical Impairments
- Bradykinesia — slowed, reduced-amplitude movement
- Akinesia — difficulty initiating movement (chair)
- Reduced right arm swing during gait
- Decreased bilateral hip and knee flexor ROM

■■ RED FLAGS

- 1. Sudden or rapid cognitive decline
- Significance: May indicate Parkinson's Disease Dementia (PDD), Lewy Body Dementia, or medication toxicity. Atypically rapid onset suggests misdiagnosis — possible Parkinson's-Plus syndrome.
- Action: Suspend physical assessment; complete MoCA cognitive screen; notify GP/neurologist urgently; do not proceed with exercise unsupervised.
- 2. Severe orthostatic hypotension or syncopal episodes
- Significance: Autonomic dysfunction in PD — a drop in systolic BP >20 mmHg on standing increases falls risk dramatically and can mimic worsening gait.
- Action: Measure lying-to-standing BP before any mobility work; cease exercise if symptomatic; refer to neurologist for medication review (levodopa and dopamine agonists can worsen hypotension).
- 3. Dysphagia or aspiration symptoms (coughing after eating/drinking)
- Significance: Bulbar involvement affects swallowing safety; aspiration pneumonia is a leading cause of mortality in PD. Even in early stages this warrants urgent attention.

■ PATHOPHYSIOLOGY (KEY POINTS)

- Parkinson's Disease is a progressive neurodegenerative disorder characterised by the selective loss of dopaminergic neurons in the substantia nigra pars compacta (SNc).
- This thalamic suppression dampens motor cortex excitation, producing the cardinal features of hypokinesia, bradykinesia, and akinesia.
- Vance's diagnosis, the disease process was likely well established for years before his resting tremor became the presenting complaint.
- Resting tremor (4–6 Hz, "pill-rolling"): Arises from pathological oscillations in the thalamo-cortical and basal ganglia-thalamic circuits when normal dampening by the dopaminergic system is lost.
- Suppressed during voluntary movement because cortical volitional drive temporarily overrides the oscillation — this is why the tremor disappears when he reaches for a cup.
- Bradykinesia and akinesia: Direct consequence of reduced dopaminergic facilitation through the direct pathway — the motor cortex receives less thalamic excitation, meaning movement initiation requires much greater cortical effort.

■ DIFFERENTIAL DIAGNOSES

See full case notes

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Key Neurological Observations (Before Formal Testing)

Tone: Passive movement of wrist and elbow bilaterally — assess for cogwheel (intermittent catches) or leadpipe (uniform) rigidity; note asymmetry (right worse expected); Froment's manoeuvre (asking patient to swing contralateral arm) can enhance subtle rigidity Coordination: Finger-nose test (absence of dysmetria distinguishes from cerebellar pathology); rapid alternating movements — pronation/supination (dysdiadochokinesia may be present due to bradykinesia, but should not have the irregular timing of cerebellar dysfunction) Posture:

Timed Up and Go (TUG)

What it measures: A composite measure of functional mobility including sit-to-stand, walking, turning (180°), walking back, and returning to sitting. It captures gait initiation speed, walking velocity, turning ability, and sit-to-stand — exactly the functional problems Mr. Vance reports. It also serves as a fall risk screening tool. How to perform: Use a standard chair with armrests (height ~46 cm); patient seated fully back in the chair Place a cone or mark on the floor exactly 3 metres in front of the chair

■ TREATMENT PLAN

- 2. "Can you describe your walking — has it changed? Your wife mentioned you trail behind her; can you tell me more about that?"
- 3. "Tell me about getting out of a chair — when it's difficult, what exactly happens? Does it feel like your body won't start moving?"
- 4. "How are your medications working? Do you notice times of day when you feel more in control of your movement — or times when it feels harder?"
- 5. "Has your handwriting changed? Can you show me how you'd sign your name or write a sentence?"
- 5. Visual hallucinations or paranoid delusions

• Significance: Can be a dopaminergic medication side effect or early indicator of Lewy Body Dementia. Indicates CNS involvement beyond the motor system.
• Action: Document exactly what the patient reports; do not reinforce the content of hallucinations; refer to neurologist for medication review; inform carer.
• Test 2: Pull Test (Postural Instability — MDS-UPDRS Item 3.12)
• What it measures:
• Patient stands with feet shoulder-width apart, eyes open, in a clear space
• Inform patient: "I'm going to give you a firm pull backwards on your shoulders. Try to stay balanced — you may take a step if you need to."
• Stand directly behind the patient, hands on both shoulders; position yourself to catch them